## Unit 5.01: Distributions & Maximum Likelihood

**Q1.** A feature in an AI vision model follows a normal distribution X sim mathcal{N}5, 4 . What is the mean mu and variance sigma^2 of this feature?

**[x] A: mu = 5, ; sigma^2 = 4**
[ ] B: mu = 4, ; sigma^2 = 5
[ ] C: mu = 5, ; sigma^2 = 2
[ ] D: mu = 2, ; sigma^2 = 4

Solution: The notation mathcal{N}mu, sigma^2 directly specifies the mean mu = 5 and variance sigma^2 = 4 with standard deviation sigma = 2 .

**Q2.** A feature in an AI vision model follows a normal distribution X sim mathcal{N}10, 9 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 10, ; sigma^2 = 3
[ ] B: mu = 9, ; sigma^2 = 10
**[x] C: mu = 10, ; sigma^2 = 9**
[ ] D: mu = 3, ; sigma^2 = 9

Solution: The notation mathcal{N}mu, sigma^2 directly specifies the mean mu = 10 and variance sigma^2 = 9 with standard deviation sigma = 3 .

**Q3.** A feature in an AI vision model follows a normal distribution X sim mathcal{N}15, 16 . What is the mean mu and variance sigma^2 of this feature?

**[x] A: mu = 15, ; sigma^2 = 16**
[ ] B: mu = 16, ; sigma^2 = 15
[ ] C: mu = 15, ; sigma^2 = 4
[ ] D: mu = 4, ; sigma^2 = 16

Solution: The notation mathcal{N}mu, sigma^2 directly specifies the mean mu = 15 and variance sigma^2 = 16 with standard deviation sigma = 4 .

**Q4.** A feature in an AI vision model follows a normal distribution X sim mathcal{N}20, 25 . What is the mean mu and variance sigma^2 of this feature?

**[x] A: mu = 20, ; sigma^2 = 25**
[ ] B: mu = 25, ; sigma^2 = 20
[ ] C: mu = 20, ; sigma^2 = 5
[ ] D: mu = 5, ; sigma^2 = 25

Solution: The notation mathcal{N}mu, sigma^2 directly specifies the mean mu = 20 and variance sigma^2 = 25 with standard deviation sigma = 5 .

**Q5.** A feature in an AI vision model follows a normal distribution X sim mathcal{N}25, 36 . What is the mean mu and variance sigma^2 of this feature?

**[x] A: mu = 25, ; sigma^2 = 36**
[ ] B: mu = 36, ; sigma^2 = 25
[ ] C: mu = 25, ; sigma^2 = 6
[ ] D: mu = 6, ; sigma^2 = 36

Solution: The notation mathcal{N}mu, sigma^2 directly specifies the mean mu = 25 and variance sigma^2 = 36 with standard deviation sigma = 6 .

**Q6.** A feature in an AI vision model follows a normal distribution X sim mathcal{N}30, 49 . What is the mean mu and variance sigma^2 of this feature?

**[x] A: mu = 30, ; sigma^2 = 49**
[ ] B: mu = 49, ; sigma^2 = 30
[ ] C: mu = 30, ; sigma^2 = 7
[ ] D: mu = 7, ; sigma^2 = 49

Solution: The notation mathcal{N}mu, sigma^2 directly specifies the mean mu = 30 and variance sigma^2 = 49 with standard deviation sigma = 7 .

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**Q7.** Given n = 4 independent observations from a normal distribution with sum sum_{i=1}^{4} x_i = 12 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**

[ ] B: hat{mu}_{text{MLE}} = 6

[ ] C: hat{mu}_{text{MLE}} = 12

[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{12}{4} = 3 .

**Q8.** Given n = 8 independent observations from a normal distribution with sum sum_{i=1}^{8} x_i = 24 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

**[x] A: hat{mu}_{text{MLE}} = 3**

[ ] B: hat{mu}_{text{MLE}} = 6

[ ] C: hat{mu}_{text{MLE}} = 24

[ ] D: hat{mu}_{text{MLE}} = 1.5

Solution: The log-likelihood ln Lmu = -frac{n}{2}ln2pisigma^2 - frac{1}{2sigma^2}sum x_i - mu^2 . Setting frac{partial ln L}{partial mu} = 0 yields the sample mean hat{mu} = frac{1}{n}sum x_i = frac{24}{8} = 3 .